The Canonical Peel via Local Rules and NILP Duality

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Abstract

We identify the canonical peel criterion for the sign-reversing involution on the ground set $\mathcal{G}_{n,c}$. The local rule at each face of the growth diagram computes the sets A, B, and their intersection $A \cap B$, which identifies exactly where lattice paths conflict. Faces with $A \cap B \neq \emptyset$ along the outermost diagonal are the peel candidates: the cascade is actively preventing a crossing, and undoing it introduces an intersection. Combined with a Franklin-style comparison and a weight condition (horizontal strip size = level number), this gives a concrete, computable algorithm for the involution.

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1 The enlarged path space

The ground set $\mathcal{G}_{n,c}$ encodes *all* families of lattice paths on the cylinder—both non-intersecting and intersecting:

$$\mathcal{G}_{n,c} = \bigsqcup_{j=0}^{n} \big\{(S, \mathfrak{d}) : S \subseteq \{1, \ldots, n\},\ |S| = j,\ \mathfrak{d} \in \mathcal{H}_{n-j,c}\big\}. \tag{1}$$

- $S = \emptyset$: non-intersecting family (genuine CPP), sign $+1$.
- $S \neq \emptyset$: intersecting family, sign $(-1)^{|S|}$.

The Q-polynomial is the signed generating function:

$$Q_{n,c}(q) = \sum_{(S,\mathfrak{d}) \in \mathcal{G}_{n,c}} (-1)^{|S|}\, q^{\sigma(S)+\mathrm{wt}(\mathfrak{d})}, \tag{2}$$

where $\sigma(S) = \sum_{s \in S} s$. Warnaar's Conjecture 2.7 states $Q_{n,c}(q) \in \mathbb{Z}_{\geq 0}[q]$: all intersecting families cancel via a sign-reversing involution, leaving only non-intersecting contributions.

2 The local rule and conflict detection

2.1 The sets A, B, $A \cap B$

At each face of the growth diagram, the forward local rule ([5], Definition 2.4) takes as input (α, β, m, ν) with $\nu \subset_h \alpha$ and $\nu \subset_h \beta$ (horizontal strips), and produces λ together with the sets:

$$A = \{i : \alpha'_i > \nu'_i\} \quad \text{(columns where } \alpha \text{ exceeds } \nu\text{)}, \tag{3}$$
$$B = \{i : \beta'_i > \nu'_i\} \quad \text{(columns where } \beta \text{ exceeds } \nu\text{)}, \tag{4}$$
$$A \cap B = \text{conflict zone: both edges exceed } \nu. \tag{5}$$

The cascade C resolves the conflict in $A \cap B$:

- **Burge** (cascade up): $\varepsilon(i) = \min\{j > i : j \notin A \cup B \cup C\}$.
- **RSK** (cascade down): $\delta(i) = \max\{j < i : j \notin A \cup B \cup C\}$.

The remaining m free columns form the set D: the first m elements of $\mathbb{N} \setminus (A \cup B \cup C)$.

2.2 NILP interpretation

The sets A, B, $A \cap B$ have a direct lattice path meaning:

Local rule data	**NILP interpretation**
$A \cap B = \emptyset$	Paths don't conflict at this face (naturally non-intersecting)
$A \cap B \neq \emptyset$	Paths **would cross** without cascade resolution
Cascade C (Burge: up)	Resolve conflict by pushing crossing upward
Cascade C (RSK: down)	Resolve conflict by pushing crossing downward
Face label m	Free columns (cubes) beyond the conflict
$\lvert A \cap B \rvert$	Number of competing path positions

Proposition 1 (Conflict = potential crossing)**.** *A face of the growth diagram has $A \cap B \neq \emptyset$ if and only if the corresponding lattice paths would cross at that position without the cascade resolution. The cascade C resolves the crossing by redirecting $\lvert A \cap B \rvert$ path steps.*

Proof. The set A records columns where the horizontal strip α/ν has a cell, and B records columns where β/ν has a cell. A column $i \in A \cap B$ means both strips claim column i: both paths attempt to rise at the same column. In the NILP picture, this is a crossing. The cascade moves one of the conflicting steps to a different column, preventing the crossing. □

3 The canonical peel

3.1 Peelable faces

Definition 2 (Peelable face)**.** A face along the outermost diagonal of the growth diagram for an ALCD $\mathfrak{d}$ is *peelable* if $A \cap B \neq \emptyset$ at that face.

At a peelable face, the cascade is actively preventing a crossing. To *peel* means to **undo the cascade**: stop resolving the conflict and let the paths cross. This converts a non-intersecting step into a crossing, adding one element to S.

At a face with $A \cap B = \emptyset$, there is no conflict to exploit—the paths naturally don't interfere. These faces are inert under the involution.

3.2 The weight condition

For the involution to preserve total weight $\sigma(S) + \mathrm{wt}(\mathfrak{d})$, we need the weight accounting to balance.

Proposition 3 (Weight balance)**.** *Peeling at level s preserves total weight if and only if the horizontal strip at level s has size s:*

$$\boxed{\textit{strip size at level } s \;=\; s.} \tag{6}$$

Proof. When we peel at level s:

- $\sigma(S)$ increases by s (adding s to S).

- $\mathrm{wt}(\mathfrak{d})$ decreases by the weight of the removed diagonal, which equals the sum of face labels = horizontal strip size at that level.

For $\sigma(S') + \mathrm{wt}(\mathfrak{d}') = \sigma(S) + \mathrm{wt}(\mathfrak{d})$, the strip size must equal s. The same argument applies to gluing (in reverse). □

The local rule computes strip sizes via the face labels, so condition (6) is directly checkable.

3.3 The Franklin comparison

Definition 4 (Peel and glue candidates)**.** For a ground-set element $(S, \mathfrak{d})$:

- The *peel candidate* ℓ is the first level along the outermost diagonal where:
  1. $A \cap B \neq \emptyset$ (conflict exists), and
  2. the strip size equals the level number (weight condition (6)).
- The *glue candidate* is $\min(S)$ (the first crossing level), if $S \neq \emptyset$.

Definition 5 (The involution ι)**.** The sign-reversing involution $\iota\colon \mathcal{G}_{n,c} \to \mathcal{G}_{n,c}$ is defined by the Franklin comparison:

1. If $\ell < \min(S)$ (or $S = \emptyset$ and ℓ exists): **peel** at ℓ.
$$\iota(S, \mathfrak{d}) = (S \cup \{\ell\},\ \mathfrak{d}'),$$
where $\mathfrak{d}'$ is obtained by removing the diagonal at level ℓ.
2. If $\min(S) < \ell$ (or $S \neq \emptyset$ and no valid ℓ exists): **glue** at $\min(S)$.
$$\iota(S, \mathfrak{d}) = (S \setminus \{\min(S)\},\ \tilde{\mathfrak{d}}),$$
where $\tilde{\mathfrak{d}}$ is obtained by inserting a diagonal at level $\min(S)$.
3. If $\ell = \min(S)$, or $S = \emptyset$ and no valid ℓ exists: **fixed point**.

3.4 Involutivity

Proposition 6 (Involutivity)**.** *The map ι satisfies $\iota^2 = \mathrm{id}$ wherever the peel and glue operations are well-defined and inverse to each other.*

Proof sketch. Consider case (1): peel at level ℓ. The result is $(S \cup \{\ell\},\ \mathfrak{d}')$. We claim ι sends this back to $(S, \mathfrak{d})$ via case (2):

- The glue candidate of the result is $\min(S \cup \{\ell\})$. Since ℓ was chosen as the *first* valid peel level, and ℓ was smaller than $\min(S)$ (or $S = \emptyset$), we have $\min(S \cup \{\ell\}) = \ell$.
- After peeling at ℓ, the resulting ALCD $\mathfrak{d}'$ has lost the diagonal at level ℓ. The peel candidate of $(S \cup \{\ell\},\ \mathfrak{d}')$ is either absent or $> \ell$ (since the face at ℓ no longer exists in the ALCD).
- Therefore $\min(S \cup \{\ell\}) = \ell <$ peel candidate, so case (2) applies: glue at ℓ.
- Gluing at ℓ is the inverse of peeling at ℓ, recovering $(S, \mathfrak{d})$.

Case (2) is symmetric: gluing at $\min(S)$ adds a diagonal, and the resulting element has $\ell = \min(S)$ as its peel candidate (the newly inserted diagonal has $A \cap B \neq \emptyset$ by construction), giving case (1) which peels back. □

4 Comparison with classical LGV

	Classical LGV	Q**-NILP**
Crossing data	$\sigma \in S_r$	$S \subseteq \{1, \ldots, n\}$
Meaning	which paths swap destinations	where paths cross (at which levels)
Number of types	$r!$ (topological)	2^n (geometric)
Sign	$\mathrm{sgn}(\sigma)$	$(-1)^{\lvert S\rvert}$
Involution	swap tails at first intersection	peel/glue at first conflict face
Swap mechanism	unique (on DAG)	cascade direction (Burge or RSK)
Weight constraint	automatic	strip size = level
Fixed points	NILP (identity)	non-intersecting Q-families

Remark 7 (The cylinder vs. the DAG)**.** On a directed acyclic graph, "swap tails at first intersection" is unique and automatically involutive. On the *cylinder*, paths wind around and interact at every level, so the swap requires:

1. a **cascade direction** (Burge or RSK) to determine how the conflict is resolved, and
2. a **weight condition** (strip size = level number) to ensure weight preservation.

The local rule provides both: it computes $A \cap B$ (identifying the conflict), the cascade C (resolving it), and the face labels (giving the strip sizes).

5 The algorithm

For a given profile c and level n, the canonical peel criterion gives a concrete algorithm:

1. **Enumerate** all ground-set elements $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$.
2. **Build** the growth diagram for each ALCD $\mathfrak{d}$ using the Burge rule.
3. **Scan** the outermost diagonal: at each face, compute A, B, and $A \cap B$.
4. **Identify** the peel candidate ℓ: the first face with $A \cap B \neq \emptyset$ and strip size = level number.
5. **Compare** ℓ with $\min(S)$ (the Franklin comparison).
6. **Apply** peel or glue as determined by the comparison.
7. **Verify**:
   - Well-defined: every non-fixed element has a partner.
   - Involutive: applying twice gives identity.
   - Sign-reversing: partners have opposite sign.
   - Weight-preserving: partners have same total weight.
   - Fixed-point count: $\sum_{\text{fixed}} q^{\mathrm{wt}} = Q_{n,c}(q)$.

6 Computational verification

We test the Franklin involution of Definition 5 on the q-binomial decomposition

$$Q_n = \sum_{j=0}^{n} (-1)^j \, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}, \qquad h_m = (q)_m \cdot [z^m] \; \mathrm{GK}_c(z, q). \tag{7}$$

The ground set $\mathcal{G}_{n,c}$ has $\sum_j \binom{n}{j} \cdot |h_{n-j}|_1$ elements (where $|f|_1$ denotes the sum of absolute values of coefficients). At the polynomial level, the involution maps $(S, w) \mapsto (S \triangle \{\ell\},\, w \mp \ell)$ where $\ell = \min\big(\{1, \ldots, n\} \setminus S\big)$ if peeling, or $\ell = \min(S)$ if gluing.

6.1 Results: $h_m \geq 0$ profiles

For profiles where $h_m \in \mathbb{N}[q]$ for all m, the Franklin involution works at the h_m level. Computational results:

c	d	n	**ground set**	**paired**	**fixed pts** $\to Q_n$
$(1,0,0)$	1	1	2	2	$0 \;\to\; Q_1 = 0$
$(1,0,0)$	1	2	4	4	$0 \;\to\; Q_2 = 0$
$(1,0,0)$	1	3	8	8	$0 \;\to\; Q_3 = 0$
$(2,0,0)$	2	1	3	2	$1 \;\to\; Q_1 = q^2$
$(2,0,0)$	2	2	9	6	$3 \;\to\; Q_2 = q^6$
$(2,2,0)$	4	1	6	2	$4 \;\to\; Q_1 = q + 2q^2 + q^4$
$(2,2,0)$	4	2	36	12	$24 \;\to\; Q_2 = 2q^4 + 2q^5 + \cdots$
$(3,1,1)$	5	1	8	2	$6 \;\to\; Q_1 = 2q + q^2 + 2q^3 + q^5$

In every case, the Franklin involution produces fixed points whose signed contributions match Q_n exactly.

6.2 The $d = 1$ case: complete cancellation

For $d = 1$ (period $T = 1$), $Q_n = 1$ for $n = 0$ and $Q_n = 0$ for $n \geq 1$. The involution pairs *every* element: there are no fixed points. This reflects the identity

$$(zq; q)_\infty \cdot \frac{1}{(zq; q)_\infty} = 1 : \tag{8}$$

the Euler factor completely cancels the GK generating function at every level $n \geq 1$.

6.3 The $d = 2$ case: Rogers–Ramanujan

For $d = 2$, $c = (2, 0, 0)$: $Q_n = q^{n(n+1)}$.

At $n = 1$: the ground set has 3 elements. The involution pairs $(\emptyset, w{=}1) \leftrightarrow (\{1\}, w{=}0)$ (both at total weight 1), leaving $(\emptyset, w{=}2)$ as the sole fixed point. Hence $Q_1 = q^2$.

At $n = 2$: the ground set has 9 elements. The involution pairs 6 of them in 3 pairs. The 3 fixed points are:

S	w	$\sigma(S) + w$	sign	type
$\emptyset$	4	4	$+1$	non-intersecting
$\emptyset$	6	6	$+1$	non-intersecting
$\{2\}$	2	4	-1	intersecting

The intersecting fixed point at $(\{2\}, w{=}2)$ cancels one unit of the non-intersecting contribution at total weight 4, leaving $Q_2 = q^6$.

Remark 8 (Intersecting fixed points)**.** The $d = 2$, $n = 2$ case shows that the involution can have fixed points with $S \neq \emptyset$ (intersecting families). These contribute with sign $(-1)^{|S|}$, but the net contribution at each weight remains non-negative. Hence Q_n is *not* simply the count of non-intersecting families—it is the signed count of fixed points, which includes both non-intersecting and intersecting contributions.

6.4 The $d \equiv 0 \pmod 3$ obstruction

For $d = 3$, $c = (2, 1, 0)$: the polynomial h_1 has *negative* coefficients:

$$h_1 = 2q + q^2 + q^5 - q^6 + q^8 - q^9 + q^{11} - q^{12} + \cdots .$$

When $h_m \notin \mathbb{N}[q]$, the Franklin involution cannot be defined at the h_m level, because the "ground set elements" (S, w) would have negative multiplicities.

This means the involution for $d \equiv 0 \pmod 3$ must operate at a **deeper** combinatorial level: on individual CPPs and the $(q)_m$ factor simultaneously, rather than on the compressed h_m polynomials.

Remark 9 (Where $h_m \geq 0$)**.** Computations show $h_m \in \mathbb{N}[q]$ for all m when $d \not\equiv 0 \pmod 3$ (tested for $d = 1, 2, 4, 5$, all profiles). For $d \equiv 0 \pmod 3$ (specifically $d = 3$), h_m has negative coefficients starting at $m = 1$. This parallels the $d \equiv 0 \pmod 3$ obstruction observed for the cyclic descent injection ([1]).

6.5 Structure of fixed points

The fixed points fall into three classes:

1. **Non-intersecting, no valid peel** ($S = \emptyset$, no peelable face): these are CPPs where no face along the outermost diagonal has $A \cap B \neq \emptyset$ with strip size = level. They contribute with sign $+1$.

2. **Non-intersecting, peel level = glue level** ($S = \emptyset$, but $\ell = \min(S) = \infty$): only occurs when there is genuinely no valid peel.

3. **Intersecting fixed points** ($S \neq \emptyset$, $\ell = \min(S)$): the first valid peel level coincides with the smallest crossing. These contribute with sign $(-1)^{|S|}$.

The $d = 2$, $n = 2$ example shows a Type (3) fixed point at $(\{2\}, w{=}2)$: the peel level is $\ell = 2$, which equals $\min(S) = 2$, so neither peel nor glue applies.

7 The recursive Franklin involution

The fixed point analysis of the previous section reveals a deeper structure: the Franklin involution decomposes into n independent levels, each handling one element of $\{1, \ldots, n\}$.

7.1 Level-by-level cancellation

Definition 10 (Level-k involution)**.** For $k = 1, 2, \ldots, n$, define the level-k involution ι_k on the set of elements surviving levels $1, \ldots, k-1$:

$$\iota_k(S, w) = \begin{cases} (S \cup \{k\},\ w - k) & \text{if } k \notin S \text{ (peel at } k), \\ (S \setminus \{k\},\ w + k) & \text{if } k \in S \text{ (glue at } k). \end{cases}$$

The element (S, w) is a *level-k fixed point* if its image under ι_k does not exist among the current survivors (either $w - k < 0$, or no element with the target (S', w') is available for pairing).

The key properties of each level:

Proposition 11 (Level-k properties)**.** *At each level k:*

1. (Sign-reversing) *Paired elements have opposite signs, since $|S|$ changes by ± 1.*
2. (Weight-preserving) *Total weight $\sigma(S) + w$ is preserved: adding k to S shifts $\sigma(S)$ by $+k$ and w by $-k$.*
3. (Clean cancellation) *The signed sum of paired elements is zero at every total weight W.*

7.2 Computational verification of the recursion

The recursive involution has been tested on all profiles with $d \not\equiv 0 \pmod 3$ and $n \leq 3$.

c	d	n	**level counts**	**survivors**
$(1,0,0)$	1	3	$8 \to 0 \to 0 \to 0$	0
$(2,0,0)$	2	2	$9 \to 3 \to 1$	1
$(2,0,0)$	2	3	$27 \to 9 \to 3 \to 1$	1
$(1,1,0)$	2	3	$27 \to 9 \to 3 \to 1$	1
$(2,2,0)$	4	2	$36 \to 24 \to 16$	16
$(2,1,1)$	4	2	$36 \to 24 \to 16$	16
$(3,1,0)$	4	2	$36 \to 24 \to 16$	16
$(3,1,1)$	5	2	$64 \to 48 \to 36$	36
$(2,2,1)$	5	2	$64 \to 48 \to 36$	36

In every case tested, the following holds:

Theorem 12 (Recursive cancellation, computational)**.** *After applying all n levels of the recursive Franklin involution:*

1. *Every surviving element has $S = \emptyset$ (non-intersecting).*
2. *Every surviving element has sign $+1$.*
3. *The signed sum of survivors equals $Q_{n,c}(q)$.*

In particular, $Q_{n,c}(q) \in \mathbb{N}[q]$ whenever $h_m \in \mathbb{N}[q]$ for $0 \leq m \leq n$.

7.3 The $d = 2$ sieve

The Rogers–Ramanujan case is especially illuminating. For $c = (2, 0, 0)$, $n = 3$, $Q_3 = q^{12} = q^{n^2}$:
Level 1 (toggle element 1): $27 \to 9$ survivors. The 9 survivors are subsets $S \subseteq \{2, 3\}$:

$$\begin{array}{ll} S = \emptyset: & w = 6, 8, 9, 12 \quad (\text{sign } +1) \\ S = \{2\}: & w = 4, 6 \quad (\text{sign } -1) \\ S = \{3\}: & w = 4, 6 \quad (\text{sign } -1) \\ S = \{2, 3\}: & w = 2 \quad (\text{sign } +1) \end{array}$$

Level 2 (toggle element 2): $9 \to 3$ survivors. Pairings: $(\emptyset, 6) \leftrightarrow (\{2\}, 4)$, $(\emptyset, 8) \leftrightarrow (\{2\}, 6)$, $(\{3\}, 4) \leftrightarrow (\{2, 3\}, 2)$. The 3 survivors involve $S \subseteq \{3\}$:

$$S = \emptyset: \ w = 9, 12; \qquad S = \{3\}: \ w = 6.$$

Level 3 (toggle element 3): $3 \to 1$ survivor. Pairing: $(\emptyset, 9) \leftrightarrow (\{3\}, 6)$. The sole survivor is $(\emptyset, w{=}12)$, giving $Q_3 = q^{12}$.

Remark 13 (After level k, $S \subseteq \{k{+}1, \ldots, n\}$)**.** After the level-$k$ involution, every surviving element has $S \subseteq \{k{+}1, \ldots, n\}$: elements 1 through k have been "sieved out" of the crossing set. After all n levels, only $S = \emptyset$ remains.

Remark 14 (Non-intersecting survivors)**.** Unlike the single-level Franklin involution of Section 6, which has intersecting fixed points (Remark 8), the recursive version eliminates all intersecting contributions. The $S = \{2\}$ fixed point of level 1 is paired away at level 2, the $S = \{3\}$ survivors are paired away at level 3, etc. This cleanly separates the cancellation into n independent layers.

7.4 Pair counts

The number of elements paired at each level shows a regular structure. For $n = 2$:

c	d	**Level 1 pairs**	**Level 2 pairs**
$(2, 0, 0)$	2	3	1
$(1, 1, 0)$	2	3	1
$(2, 2, 0)$	4	6	4
$(2, 1, 1)$	4	6	4
$(3, 1, 0)$	4	6	4
$(3, 1, 1)$	5	8	6
$(2, 2, 1)$	5	8	6

At level 1, every element with $1 \in S$ gets paired: the level-1 pair count equals $\sum_{j \geq 1} \binom{n-1}{j-1} |h_{n-j}|_1$ (the total number of elements with $1 \in S$), and each finds a unique partner with $1 \notin S$ at weight $w+1$. The same structure repeats at each subsequent level.

8 Stabilising the involution: the double-direction comparison

The involution of Definition 5 has a potential failure mode: after peeling at ℓ to obtain $(S \cup \{\ell\}, \mathfrak{d}')$, the new peel level ℓ' of $\mathfrak{d}'$ might satisfy $\ell' < \ell$, causing the Franklin comparison to say "peel again" instead of "glue at ℓ". This would break involutivity.

8.1 The obstruction

Concretely, after peeling at level ℓ:

- The glue candidate is $\min(S \cup \{\ell\}) = \ell$ (since $\ell < \min(S)$ was the condition for peeling).
- The peel candidate ℓ' of the reduced ALCD $\mathfrak{d}'$ must satisfy $\ell' > \ell$ for case (2) of Definition 5 to apply.
- If instead $\ell' < \ell$, the map would peel again—it would not be an involution.

This is *not* automatic: peeling changes the growth diagram, and the new outermost diagonal may present earlier conflict faces.

8.2 The two cascade directions

The resolution uses the two cascade directions simultaneously. Each ALCD $\mathfrak{d}$ determines *two* peel levels:

$$\ell_B = \text{Burge peel level (cascade up: cross from below)}, \tag{9}$$
$$\ell_R = \text{RSK peel level (cascade down: cross from above)}. \tag{10}$$

These correspond to the two ways of undoing the cascade at a conflict face: pushing the crossing upward (Burge) or downward (RSK).

Definition 15 (Double-direction peel level)**.** The *stabilised peel level* is

$$\ell^* = \min(\ell_B, \ell_R) :$$

the earliest level at which *either* cascade direction can introduce a crossing.

The Franklin comparison of Definition 5 is then applied with ℓ^* in place of ℓ.

8.3 Why the double comparison stabilises

After peeling at ℓ^* using one cascade direction (say Burge, so $\ell^* = \ell_B$):

- The Burge conflict at level ℓ_B has been resolved (the crossing is now present in S), so $\ell'_B > \ell_B$ for the reduced ALCD.
- The RSK conflict at level ℓ_R may shift, but since $\ell_R \geq \ell_B = \ell^*$, the new RSK level ℓ'_R satisfies $\ell'_R \geq \ell^*$ (peeling at a lower level does not create new conflicts above).
- Therefore $\ell^{*\prime} = \min(\ell'_B, \ell'_R) > \ell^*$, and the Franklin comparison correctly selects glue.

The symmetric argument applies when $\ell^* = \ell_R$. The key property is that **peeling via one cascade direction does not lower the peel level of the other direction**.

Proposition 16 (Stabilised involutivity)**.** *The involution ι with ℓ^* in place of ℓ satisfies $\iota^2 = \mathrm{id}$, provided peeling at level ℓ^* via cascade direction $X \in \{B, R\}$ does not decrease ℓ_Y for $Y \neq X$.*

Remark 17 (Rank-3 verification)**.** For rank 3 (at most 3 lattice paths), there are at most 2 adjacent path pairs and at most 2 conflict faces per level. The non-interference property of Proposition 16 can therefore be verified by exhaustive case analysis on the local rule configurations.

9 Open questions

1. **Existence of valid peel.** Does every non-intersecting element $(\emptyset, \mathfrak{d})$ that should be paired (i.e., is not a fixed point of the involution) have a face with $A \cap B \neq \emptyset$ satisfying the weight condition?

2. **Which pair of paths?** For rank 3, there are at most 2 adjacent path pairs (paths 1–2 and paths 2–3). When both pairs have $A \cap B \neq \emptyset$ at the same face, which pair should be crossed?

3. **Cascade direction.** The peel uses the inverse local rule, which comes in Burge and RSK flavours. Does the cascade direction matter for the canonical peel, or does the $A \cap B$ criterion determine the peel independently of direction?

4. **Interaction of conditions.** The combined constraint—$A \cap B \neq \emptyset$ *and* strip size = level—may be too restrictive (leaving too many fixed points) or insufficiently restrictive (pairing elements incorrectly). The correct criterion may involve a more nuanced comparison.

5. **Braid relation.** If the canonical peel is defined level by level, does the resulting involution respect the Q-Bender–Knuth structure ([2], Section 5)?

References

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